

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 1 EXAMINATION 2024-2025****IE4476 – IMAGE PROCESSING & COMPUTER VISION**

November / December 2024

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a Restricted Open Book Examination. You are allowed to bring into the examination hall one double-sided A4-size reference sheet with texts and figures handwritten, drawn, or printed on the A4 paper (no sticky notes/post-it notes on the reference sheet).
5. Unless specifically stated, all symbols have their usual meanings.

1. During an imaging process, a continuous image $f(x, y)$ undergoes uniform planar motion in the x- and y-directions, given as

$$x_0(t) = 0 \quad \text{and} \quad y_0(t) = t \quad \text{for} \quad 0 \leq t < 1,$$

resulting in a degraded image $g(x, y)$.

- (a) Express the degraded image $g(x, y)$ in terms of the original image $f(x, y)$.
(10 Marks)
- (b) Assuming there is no other form of degradation, design a filter to restore the original image $f(x, y)$ from the degraded image $g(x, y)$.
(8 Marks)
- (c) If image $g(x, y)$ is also degraded by additive noise which has zero mean and is uncorrelated to the original image $f(x, y)$, design a filter to restore the original image $f(x, y)$ from the degraded image $g(x, y)$. Clearly state your assumption(s), if any.
(7 Marks)

2. Consider a student with a height of 1.67 m. You need to judge whether the student is male or female.
- (a) Suppose you only know that there are 300 male students and 700 female students. What is your decision based on this knowledge? Justify that your decision is optimal. (5 Marks)
- (b) Suppose you only know that the average height of male students is 1.7 m and that of female students is 1.62 m. What is your decision based on this knowledge? Under what conditions (assumptions) can your decision be considered optimal? Clearly justify your answer. (10 Marks)
- (c) Suppose you know both the information in parts (a) and (b) and you also know that the standard deviations of the heights of the male and female students are 0.2 m and 0.3 m, respectively. What is your decision and under what conditions (assumptions) your decision is optimal? (10 Marks)
3. A linear transform $\mathbf{W}$ is applied to transform feature vector $\mathbf{x}$ into feature vector $\mathbf{y}$, given as $\mathbf{y} = \mathbf{W}^T \mathbf{x}$.
- (a) Express the Euclidean distance between two linearly transformed feature vectors $d_u(\mathbf{y}_1, \mathbf{y}_2)$ in terms of the two corresponding feature vectors $\mathbf{x}_1, \mathbf{x}_2$. Derive the condition(s) under which $d_u(\mathbf{y}_1, \mathbf{y}_2) = d_u(\mathbf{x}_1, \mathbf{x}_2)$ for any $\mathbf{x}_1, \mathbf{x}_2$. (12 Marks)
- (b) The Mahalanobis distance is defined as $d_m(\mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_1 - \mathbf{x}_2)^T \Sigma_x^{-1} (\mathbf{x}_1 - \mathbf{x}_2)$ where Σ_x is the covariance matrix of the feature vector $\mathbf{x}$. Express the Mahalanobis distance $d_m(\mathbf{y}_1, \mathbf{y}_2)$ in terms of $\mathbf{x}_1, \mathbf{x}_2$ and derive the condition(s) under which $d_m(\mathbf{y}_1, \mathbf{y}_2) = d_m(\mathbf{x}_1, \mathbf{x}_2)$ for any $\mathbf{x}_1, \mathbf{x}_2$. Compare the results with those in part (a). (13 Marks)

4. (a) Let $\mathbf{x} = [x_1, x_2, \dots, x_{100}]^T$ and $\mathbf{y} = [y_1, y_2, \dots, y_{98}]^T$ denote, respectively, the inputs and the outputs of a fully-connected layer of a neural network with activation function $s(\cdot)$. The trainable network parameters to connect the inputs and outputs are given by $\mathbf{W} = [w_{ij}]$ and $\boldsymbol{\theta} = [\theta_1, \theta_1, \dots, \theta_{98}]^T$, where the matrix $\mathbf{W} = [w_{ij}]$ has 98 rows and 100 columns.

- (i) Express the outputs in term of inputs in vector-matrix form and scalar form.
- (ii) Compute the number of trainable parameters and the numbers of multiplications and summations required to compute the outputs from the inputs of this fully-connected layer.
- (iii) What is the ratio of the number of outputs to the number of trainable parameters?

(12 Marks)

- (b) Let $\mathbf{x} = [x_1, x_2, \dots, x_{100}]^T$ and $\mathbf{y}^k = [y_2^k, y_3^k, \dots, y_{99}^k]^T$, $1 \leq k \leq 20$, denote the inputs and the outputs, respectively, of a convolutional layer of a neural network with activation function $r(\cdot)$. The convolutional layer has 20 learnable filters of size 3 represented by $\mathbf{w}^k = [w_{-1}^k, w_0^k, w_1^k]^T$ and bias θ^k , $1 \leq k \leq 20$.

- (i) Express the outputs in term of inputs in scalar form.
- (ii) Compute the number of trainable parameters and the numbers of multiplications and summations required to compute the outputs from the inputs of this convolutional layer.
- (iii) What is the ratio of the number of outputs to the number of trainable parameters?

(13 Marks)

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.